

1.

$$a) P(y|\mu) = e^{-\mu} \mu^y / y!$$

$$L(\mu|y) = \prod_{i=1}^n e^{-\mu_i} \mu_i^{y_i} / y_i!$$

$$L(\beta|y) = \prod_{i=1}^n e^{-e^{x_i^T \beta}} (e^{x_i^T \beta})^{y_i} / y_i!$$

$$J(\beta) = -\log L(\beta)$$

$$= -\log \left(\prod_{i=1}^n e^{-e^{x_i^T \beta}} (e^{x_i^T \beta})^{y_i} / y_i! \right)$$

$$= - \sum_{i=1}^n \log(e^{-e^{x_i^T \beta}}) + \log((e^{x_i^T \beta})^{y_i}) - \log(y_i!)$$

$$= - \sum_{i=1}^n -e^{x_i^T \beta} + y_i (x_i^T \beta) - \log(y_i!)$$

$$= \sum_{i=1}^n \log(y_i!) + e^{x_i^T \beta} - y_i (x_i^T \beta)$$

$$\approx \sum_{i=1}^n e^{x_i^T \beta} - y_i (x_i^T \beta)$$

(discarding terms not depending on β) •

$$b) J(\beta) = \sum_{i=1}^n e^{x_i^T \beta} - y_i (x_i^T \beta)$$

$$\nabla J(\beta) = \frac{dJ(\beta)}{d\beta} = \begin{bmatrix} \frac{dJ(\beta)}{d\beta_1} \\ \vdots \\ \frac{dJ(\beta)}{d\beta_0} \end{bmatrix}$$

→ we will try to find a common pattern to evaluate $\frac{dJ(\beta)}{d\beta_j}$

We know $J(\beta) = \sum_{i=1}^n e^{x_{i1}\beta_1} e^{x_{i2}\beta_2} \dots e^{x_{i0}\beta_0} - y_i (x_{i1}\beta_1 + \dots + x_{i0}\beta_0)$

↓
as per chain rule,

$$\frac{d(e^{x_{i1}\beta_1} e^{x_{i2}\beta_2} \dots e^{x_{i0}\beta_0})}{d\beta_j} = \frac{d(y_i (x_{i1}\beta_1 + \dots + x_{i0}\beta_0))}{d\beta_j} = x_{ij} y_i$$

$$= x_{ij} e^{x_i^T \beta}$$

$$\therefore \nabla J(\beta) = \begin{bmatrix} \sum_{i=1}^n x_{i1} e^{x_i^T \beta} - x_{i1} y_i \\ \vdots \\ \sum_{i=1}^n x_{i0} e^{x_i^T \beta} - x_{i0} y_i \end{bmatrix}$$

$$= \begin{bmatrix} \sum_{i=1}^n x_{i1} (e^{x_i^T \beta} - y_i) \\ \vdots \\ \sum_{i=1}^n x_{i0} (e^{x_i^T \beta} - y_i) \end{bmatrix}$$

In words, each entry of $\nabla J(\beta)$ is the dot product between the j th column of x (x_{ij}) and the difference between the estimated conditional Poisson mean ($e^{x_i^T \beta}$) and y_i .

1.

$$c) \mathcal{J}(\beta) = \begin{bmatrix} \frac{d^2 J(\beta)}{d\beta_1^2} & \frac{d^2 J(\beta)}{d\beta_2 d\beta_1} & & \\ \frac{d^2 J(\beta)}{d\beta_1 d\beta_2} & & & \\ & & \ddots & \\ & & & \frac{d^2 J(\beta)}{d\beta_{10}^2} \end{bmatrix}$$

We must find a pattern for $\frac{d^2 J(\beta)}{d\beta_j d\beta_k}$.

We know $\frac{dJ(\beta)}{d\beta_j} = \sum_{i=1}^n x_{ij} (e^{x_i^T \beta} - y_i)$ as per b)

It follows that $\frac{d^2 J(\beta)}{d\beta_j d\beta_k} = \sum_{i=1}^n x_{ij} x_{ik} e^{x_i^T \beta}$
 $= \sum_{i=1}^n x_{ij} x_{ik} \mu_i$ (for the sake of clarity)

$$J(\beta) = X^T W X \Rightarrow \begin{bmatrix} \sum x_{i1}^2 \mu_i & \sum x_{i1} x_{i2} \mu_i & & \\ \sum x_{i2} x_{i1} \mu_i & & & \\ & & \ddots & \\ & & & \sum x_{i10}^2 \mu_i \end{bmatrix} = \begin{bmatrix} x_{11} & x_{12} & & \\ x_{21} & & & \\ & & \ddots & \\ & & & x_{10n} \end{bmatrix}^T W \begin{bmatrix} x_{11} & x_{12} & & \\ x_{21} & & & \\ & & \ddots & \\ & & & x_{10n} \end{bmatrix}$$

W must be a square matrix of dimension $n \times n$ to produce a valid Hessian matrix of dimension 10×10 . Knowing that $(X^T X)_{jk} = \sum_{i=1}^{10} x_{ij} x_{ik}$, we need W to multiply each term by μ_i within the sum, while discarding any term that isn't included. The simplest matrix that achieves this is $W = \text{diag}(\mu_1, \dots, \mu_n)$, effectively performing row-wise multiplication as mentioned before.

$$\therefore J(\beta) = X^T \begin{bmatrix} \mu_1 & 0 & 0 & \dots & 0 \\ 0 & \mu_2 & & & \\ 0 & & \ddots & & \\ 0 & & & \ddots & \\ 0 & & & & \mu_n \end{bmatrix} X = X^T \begin{bmatrix} e^{x_1^T \beta} & 0 & \dots & 0 \\ 0 & e^{x_2^T \beta} & & \\ \vdots & & \ddots & \\ 0 & & & e^{x_n^T \beta} \end{bmatrix} X$$

$$\begin{aligned}
 5. \quad b) \quad H_n(\beta) &= E \left[\nabla^2 J(\beta) \right] \\
 &= E \left[X^T W X \right] \quad (\text{as per 1c.)}) \\
 &= \begin{bmatrix} E[\sum x_{i1}^2 \mu_i] & E[\sum x_{i1} x_{i2} \mu_i] & \dots & E[\sum x_{i1} x_{in} \mu_i] \\ \vdots & \ddots & \ddots & \vdots \\ E[\sum x_{in} x_{i1} \mu_i] & E[\sum x_{i2} x_{i1} \mu_i] & \dots & E[\sum x_{in}^2 \mu_i] \end{bmatrix} \\
 &\quad \text{by linearity of expectation}
 \end{aligned}$$

But, since X is known and W is derived from X and β ($W = \text{diag}(\mu)$), $E[X^T W X] = X^T W X$.

$$G_n(\beta) = E \left[s(\beta) s(\beta)^T \right] \quad \left(s(\beta) = \sum_{i=1}^n x_i (\mu_i - y_i) \right)$$

$$= E \left[\sum_{i=1}^n \sum_{j=1}^n x_i (\mu_i - y_i) (\mu_j - y_j) x_j^T \right]$$

$$= \sum_{i=1}^n \sum_{j=1}^n x_i x_j^T E[(\mu_i - y_i)(\mu_j - y_j)]$$

by linearity of expectation, and given X is fixed.

We have two cases:

① $i=j$:

$$\text{Term is } x_i x_i^T E[(\mu_i - y_i)^2]$$

$$= x_i x_i^T V[y_i | x_i]$$

② $i \neq j$:

$$\text{Term is } x_i x_j^T E[(\mu_i - y_i)(\mu_j - y_j)]$$

$$= x_i x_j^T E[\mu_i - y_i] E[\mu_j - y_j]$$

as rows are independent.

Under correct model conditional mean specification, we have $E[y_i | x_i] = \mu_i$

$$\Rightarrow \text{Term is } x_i x_j^T \cdot 0 \cdot 0$$

$$= 0$$

$$\therefore G_n(\beta) = \sum_{i=1}^n x_i x_i^T V(y_i | x_i) = X^T \text{diag}(V) X$$

If the Poisson specification is correct, then $V(y_i | x_i) = E[y_i | x_i] = \mu_i$

$$\begin{aligned}
 \Rightarrow G_n(\beta) &= \sum_{i=1}^n x_i x_i^T \mu_i = X^T \text{diag}(\mu) X \\
 &= X^T W X \\
 &= H_n(\beta)
 \end{aligned}$$

5.

c) With row independence + correct conditional mean model, cross-terms are removed (as per 5b)):

$$G_n(\beta) = \sum_{i=1}^n x_i x_i^T E[(\mu_i - y_i)^2]$$

$$= \sum_{i=1}^n x_i x_i^T V[y_i | x_i]$$

$$= \phi \sum x_i x_i^T \mu_i$$

$$= \phi H_n(\beta) \quad (H_n(\beta) \text{ derived in 5b))}$$

Under a robust approach, using the sandwich estimator:

$$\text{Var}(\hat{\beta}) = H_n^{-1} G_n H_n^{-1}$$

$$= H_n^{-1} (\phi H_n) H_n^{-1}$$

$$= \phi H_n^{-1} \Rightarrow \hat{se}_\beta = \sqrt{\phi} \hat{se}_{\text{model}}$$